367 Old State Rd
Berwyn, PA, 19312

Hansen Yi

484-787-4906 | hansenyi@seas.upenn.edu | www.linkedin.com/in/hansen-yi | hansen-yi.github.io/

EDUCATION

University of Pennsylvania | Philadelphia, PA — May 2024

Master of Science in Engineering in Computer and Information Science
Bachelor of Science in Engineering in Digital Media Design (Computer Science), summa cum laude
Cumulative GPA: 3.81/4.00; Dean's List
Relevant Coursework: Proving Things: Analysis; Probability; Mathematical Foundations of Computer Science; Introduction to Computer Systems; Programming Languages and Techniques I; Data Structures and Algorithms; Calculus III; Principles of Physics II: Electromagnetism and Radiation; 3-D Computer Modeling; Interactive Computer Graphics; Architecture; Automata, Computability, and Complexity; Introduction to Algorithms; Computer Animation; Artificial Intelligence; Advanced Rendering; Game Design Practicum; Big Data Analytics; Database and Information Systems; Computer Organization and Design; Analysis of Algorithms; Networked Systems; Procedural Computer Graphics; Introduction to Human Computer Interaction

PROFESSIONAL EXPERIENCE

University of Pennsylvania, *Advanced Rendering Teaching Assistant* | Philadelphia, PA — February 2024 – May 2024

- Conducted office hours to aid students in debugging C++ and GLSL code, drawing from my experience developing a Monte Carlo path tracer that employs multiple importance sampling, direct, indirect and environment lighting during my coursework
- Graded homework and quizzes, answered students' questions, and participated in meetings regarding the course's direction

Ben Franklin AI Research Project, *CIS Research Assistant (Software Developer)* | Philadelphia, PA — July 2023 – September 2023

- Developed a virtual representation of Benjamin Franklin using Unreal Engine's MetaHumans, incorporating facial tracking technology, custom hair design, photogrammetry, and AI powered tools for voice modulation and facial animation generation

Engineering Summer Academy at Penn, *Computer Graphics RTA* | Philadelphia, PA — July 2022

- Collaborated with 2 other RTAs to provide guidance to students in creating intricate character models using Autodesk Maya
- Orchestrated engaging activities, prioritized student safety in and out of class, fostering an immersive learning environment

SELECTED PROJECTS

Newscast Mobile App (*React Native, Expo Go, Python, JavaScript – Team of 3*) — June 2023 – August 2023

- Leveraged Git for streamlined virtual collaboration and task management, ensuring efficient teamwork and role tracking
- Orchestrated the frontend development according to provided designs, integrating third-party libraries for the audio player
- Established seamless communication with the backend using HTTP requests, retrieving data from MongoDB GridFS buckets

Restaurant Hunter (*SQL, HTML, CSS, JavaScript, React – Team of 4*) — May 2023

- Crafted a dynamic web application with a React-powered frontend and a backend driven by JavaScript and SQL
- Enhanced database performance by optimizing SQL queries, enabling faster data retrieval from a substantial Yelp dataset
- Integrated user-defined parameters with the OpenLayers API, offering an interactive map display of restaurant locations

Superscalar Pipelined Processor (*Verilog*) — April 2023

- Designed a fully bypassed superscalar pipelined LC4 processor with two in-order 5-stage pipelines featuring branch prediction, data forwarding, stalling, pipe-switching, and hazard detection, as well as a multi-ported bypassed register file

Hungry Bunnies (*Unreal Engine 5, C++ – Team of 3*) — March 2023

- Implemented game logic using Blueprints including collisions mechanics, player conflict interactions, and a scoring system
- Conceived and developed multiple iterations of the game, iteratively refining it based on rigorous playtesting feedback

Mini Minecraft (*C++, GLSL – Team of 3*) — April 2022

- Collaborated to develop an immersive, simpler version of Minecraft, leveraging noise functions to generate and interpolate several different procedural terrains, and created post-processing shaders using the OpenGL Pipeline

Mini Maya (*C++*) — March 2022

- Constructed a simple version of Maya with the ability to import and edit mesh structures, and load and bind skeleton joints

J Compiler (*C*) — December 2021

- Developed a compiler to convert J, a stack-oriented language, into LC4 assembly code emulating the lcc compiler for C

Tetris (*Java, Swing*) — April 2021

- Developed a feature-rich Tetris game, implementing core game mechanics using a variety of data structures
- Implemented progress saving functionality with File I/O, and streamlined debugging with comprehensive JUnit test cases

SKILLS

Technical: Java, C++, Python, C#, SQL, GLSL, JavaScript, TypeScript, Three.js, HTML, CSS, Git, C, Verilog, React Native, OCaml, Assembly, LaTeX, MongoDB, Neo4j, React, Pandas, Apache Spark, NumPy, Sklearn, Seaborn, NLTK, PyTorch, AWS, Unity, Software Engineering, Unreal Engine, Houdini, Adobe Creative Cloud, Rhino 7, Autodesk Maya, MotionBuilder, Microsoft Office
Language: Mandarin (Fluent), Spanish (Intermediate), Japanese (Beginner)